

CHASSIS C1 DI/DO FACTORY

Project: SPRINT 18K Spray Dryer | Target

TOTAL CHANNELS	PASSED CHANNELS
192	192 (100%)
64 DO + 128 DI Channels	Zero Faults Detected

1. CHASSIS C1 HARDWARE CONFIGURATION & TEST COVERAGE

Slot #	Module Model	Description / Function	Channel Count
C1S0	1756-L950TPSXT	ControlLogix 5580 Master Processor (20MB)	1x CPU
C1S1	1756-EN4TR	EtherNet/IP 1Gbps DLR Supervisor Port	2x DLR Ports
C1S2	1756-EN4TR	EtherNet/IP 1Gbps Plant Network Uplink	2x Ports
C1S3	1756-EN4TR	EtherNet/IP 1Gbps SCADA Communication	2x Ports
C1S4	1756-IB32	24VDC Digital Input Module (DI Group 1)	32 Channels
C1S5	1756-IB32	24VDC Digital Input Module (DI Group 2)	32 Channels
C1S6	1756-IB32	24VDC Digital Input Module (DI Group 3)	32 Channels
C1S7	1756-IB32	24VDC Digital Input Module (DI Group 4)	32 Channels
C1S8	1756-OB32	24VDC Digital Output Module (DO Group 1)	32 Channels
C1S9	1756-OB32	24VDC Digital Output Module (DO Group 2)	32 Channels
C1S10	1756-IF16	Analog Input 4-20mA (Slurry & Spray Feed)	16 Channels
C1S11	1756-IF16	Analog Input 4-20mA (Burner & Chamber Temp)	16 Channels
C1S12	1756-N2	Unpopulated Slot Position	-

ACCEPTANCE TEST (FAT) MASTER REPORT

: MCP-01 Chassis C1 | Document: KAL-FAT-TR-C1-2608 | Rev 1.0

FAILED CHANNELS	AVG SWITCH I
0 (0.0%)	28.9 n
100% Signal Integrity	Standard Spec: ·

Field Destination	FAT Test Methodology	Test Status
Central Area (CA1)	Firmware v34, memory & CPU diagnostics	100% Verified
DLR Ring & Cisco 9300	Ring break recovery test (<3ms)	100% Verified
Plant Core Backbone	VLAN tagging & HSRP ping test	100% Verified
FactoryTalk View SE Server	SCADA poll rate & packet latency check	100% Verified
CA1, JB-401 (TBDI1)	Closed-Loop Cross-Wiring from C1S8 Relays	32/32 Tested
JB-401 (TBDI2)	Closed-Loop Cross-Wiring from C1S9 Relays	32/32 Tested
JB-401, JB-601 (TBDI3)	Direct 24VDC Dry Contact Simulation	32/32 Tested
JB-601 (TBDI4)	Direct 24VDC Dry Contact Simulation	32/32 Tested
Relays P1-TBRL1 (1-32)	Force DO -> Relay A1 -> NO 11/14 -> C1S4	32/32 Tested
Relays P1-TBRL2 (1-32)	Force DO -> Relay A1 -> NO 11/14 -> C1S5	32/32 Tested
JB-401 (TBAI1)	Differential 5-Point Calibration (Fluke 789)	16/16 Tested
JB-601 (TBAI2)	Differential 5-Point Calibration (Fluke 789)	16/16 Tested
Spare Slot	Slot Cover Installed	N/A



LATENCY

ns

 $< 200 \text{ ms}$ [illegible]

CHASSIS C1 — DI/DO CL

Verification: C1S8 DO -> P1-TBRL

Loop #	DO Slot & Pin	Interposing Relay Coil	Relay Contact NO
C1-LPB-001	C1S8.1	P1-TBRL-1 (A1/A2)	P1-TBRL-1 (11/14)
C1-LPB-002	C1S8.2	P1-TBRL-2 (A1/A2)	P1-TBRL-2 (11/14)
C1-LPB-003	C1S8.3	P1-TBRL-3 (A1/A2)	P1-TBRL-3 (11/14)
C1-LPB-004	C1S8.4	P1-TBRL-4 (A1/A2)	P1-TBRL-4 (11/14)
C1-LPB-005	C1S8.5	P1-TBRL-5 (A1/A2)	P1-TBRL-5 (11/14)
C1-LPB-006	C1S8.6	P1-TBRL-6 (A1/A2)	P1-TBRL-6 (11/14)
C1-LPB-007	C1S8.7	P1-TBRL-7 (A1/A2)	P1-TBRL-7 (11/14)
C1-LPB-008	C1S8.8	P1-TBRL-8 (A1/A2)	P1-TBRL-8 (11/14)
C1-LPB-009	C1S8.9	P1-TBRL-9 (A1/A2)	P1-TBRL-9 (11/14)
C1-LPB-010	C1S8.10	P1-TBRL-10 (A1/A2)	P1-TBRL-10 (11/14)
C1-LPB-011	C1S8.11	P1-TBRL-11 (A1/A2)	P1-TBRL-11 (11/14)
C1-LPB-012	C1S8.12	P1-TBRL-12 (A1/A2)	P1-TBRL-12 (11/14)
C1-LPB-013	C1S8.13	P1-TBRL-13 (A1/A2)	P1-TBRL-13 (11/14)
C1-LPB-014	C1S8.14	P1-TBRL-14 (A1/A2)	P1-TBRL-14 (11/14)
C1-LPB-015	C1S8.15	P1-TBRL-15 (A1/A2)	P1-TBRL-15 (11/14)
C1-LPB-016	C1S8.16	P1-TBRL-16 (A1/A2)	P1-TBRL-16 (11/14)
C1-LPB-017	C1S8.19	P1-TBRL-17 (A1/A2)	P1-TBRL-17 (11/14)
C1-LPB-018	C1S8.20	P1-TBRL-18 (A1/A2)	P1-TBRL-18 (11/14)
C1-LPB-019	C1S8.21	P1-TBRL-19 (A1/A2)	P1-TBRL-19 (11/14)
C1-LPB-020	C1S8.22	P1-TBRL-20 (A1/A2)	P1-TBRL-20 (11/14)
C1-LPB-021	C1S8.23	P1-TBRL-21 (A1/A2)	P1-TBRL-21 (11/14)
C1-LPB-022	C1S8.24	P1-TBRL-22 (A1/A2)	P1-TBRL-22 (11/14)
C1-LPB-023	C1S8.25	P1-TBRL-23 (A1/A2)	P1-TBRL-23 (11/14)
C1-LPB-024	C1S8.26	P1-TBRL-24 (A1/A2)	P1-TBRL-24 (11/14)
C1-LPB-025	C1S8.27	P1-TBRL-25 (A1/A2)	P1-TBRL-25 (11/14)
C1-LPB-026	C1S8.28	P1-TBRL-26 (A1/A2)	P1-TBRL-26 (11/14)
C1-LPB-027	C1S8.29	P1-TBRL-27 (A1/A2)	P1-TBRL-27 (11/14)

C1-LPB-028	C1S8.30	P1-TBRL-28 (A1/A2)	P1-TBRL-28 (11/14)
C1-LPB-029	C1S8.31	P1-TBRL-29 (A1/A2)	P1-TBRL-29 (11/14)
C1-LPB-030	C1S8.32	P1-TBRL-30 (A1/A2)	P1-TBRL-30 (11/14)
C1-LPB-031	C1S8.33	P1-TBRL-31 (A1/A2)	P1-TBRL-31 (11/14)
C1-LPB-032	C1S8.34	P1-TBRL-32 (A1/A2)	P1-TBRL-32 (11/14)
C1-LPB-033	C1S9.1	P1-TBRL-33 (A1/A2)	P1-TBRL-33 (11/14)
C1-LPB-034	C1S9.2	P1-TBRL-34 (A1/A2)	P1-TBRL-34 (11/14)
C1-LPB-035	C1S9.3	P1-TBRL-35 (A1/A2)	P1-TBRL-35 (11/14)
C1-LPB-036	C1S9.4	P1-TBRL-36 (A1/A2)	P1-TBRL-36 (11/14)
C1-LPB-037	C1S9.5	P1-TBRL-37 (A1/A2)	P1-TBRL-37 (11/14)
C1-LPB-038	C1S9.6	P1-TBRL-38 (A1/A2)	P1-TBRL-38 (11/14)
C1-LPB-039	C1S9.7	P1-TBRL-39 (A1/A2)	P1-TBRL-39 (11/14)
C1-LPB-040	C1S9.8	P1-TBRL-40 (A1/A2)	P1-TBRL-40 (11/14)
C1-LPB-041	C1S9.9	P1-TBRL-41 (A1/A2)	P1-TBRL-41 (11/14)
C1-LPB-042	C1S9.10	P1-TBRL-42 (A1/A2)	P1-TBRL-42 (11/14)
C1-LPB-043	C1S9.11	P1-TBRL-43 (A1/A2)	P1-TBRL-43 (11/14)
C1-LPB-044	C1S9.12	P1-TBRL-44 (A1/A2)	P1-TBRL-44 (11/14)
C1-LPB-045	C1S9.13	P1-TBRL-45 (A1/A2)	P1-TBRL-45 (11/14)
C1-LPB-046	C1S9.14	P1-TBRL-46 (A1/A2)	P1-TBRL-46 (11/14)
C1-LPB-047	C1S9.15	P1-TBRL-47 (A1/A2)	P1-TBRL-47 (11/14)
C1-LPB-048	C1S9.16	P1-TBRL-48 (A1/A2)	P1-TBRL-48 (11/14)
C1-LPB-049	C1S9.19	P1-TBRL-49 (A1/A2)	P1-TBRL-49 (11/14)
C1-LPB-050	C1S9.20	P1-TBRL-50 (A1/A2)	P1-TBRL-50 (11/14)
C1-LPB-051	C1S9.21	P1-TBRL-51 (A1/A2)	P1-TBRL-51 (11/14)
C1-LPB-052	C1S9.22	P1-TBRL-52 (A1/A2)	P1-TBRL-52 (11/14)
C1-LPB-053	C1S9.23	P1-TBRL-53 (A1/A2)	P1-TBRL-53 (11/14)
C1-LPB-054	C1S9.24	P1-TBRL-54 (A1/A2)	P1-TBRL-54 (11/14)
C1-LPB-055	C1S9.25	P1-TBRL-55 (A1/A2)	P1-TBRL-55 (11/14)
C1-LPB-056	C1S9.26	P1-TBRL-56 (A1/A2)	P1-TBRL-56 (11/14)
C1-LPB-057	C1S9.27	P1-TBRL-57 (A1/A2)	P1-TBRL-57 (11/14)
C1-LPB-058	C1S9.28	P1-TBRL-58 (A1/A2)	P1-TBRL-58 (11/14)
C1-LPB-059	C1S9.29	P1-TBRL-59 (A1/A2)	P1-TBRL-59 (11/14)

C1-LPB-060	C1S9.30	P1-TBRL-60 (A1/A2)	P1-TBRL-60 (11/14)
C1-LPB-061	C1S9.31	P1-TBRL-61 (A1/A2)	P1-TBRL-61 (11/14)
C1-LPB-062	C1S9.32	P1-TBRL-62 (A1/A2)	P1-TBRL-62 (11/14)
C1-LPB-063	C1S9.33	P1-TBRL-63 (A1/A2)	P1-TBRL-63 (11/14)
C1-LPB-064	C1S9.34	P1-TBRL-64 (A1/A2)	P1-TBRL-64 (11/14)

POSED-LOOP CROSS-WIRING VERIFICATION (64 LOOPS)

1 Relay -> C1S4 DI (Loops 1-32) | C1S9 DO -> P1-TBRL2 Relay -> C1S5 DI (Loops 33-64)

Temp Jumper Wire	Target DI Terminal	DI Slot & Pin	Switch Latency
TMP-JMP-001	P1-TBDI-1 (Pin 1)	C1S4.1	40 ms
TMP-JMP-002	P1-TBDI-2 (Pin 2)	C1S4.2	28 ms
TMP-JMP-003	P1-TBDI-3 (Pin 3)	C1S4.3	39 ms
TMP-JMP-004	P1-TBDI-4 (Pin 4)	C1S4.4	33 ms
TMP-JMP-005	P1-TBDI-5 (Pin 5)	C1S4.5	36 ms
TMP-JMP-006	P1-TBDI-6 (Pin 6)	C1S4.6	23 ms
TMP-JMP-007	P1-TBDI-7 (Pin 7)	C1S4.7	38 ms
TMP-JMP-008	P1-TBDI-8 (Pin 8)	C1S4.8	28 ms
TMP-JMP-009	P1-TBDI-9 (Pin 9)	C1S4.9	41 ms
TMP-JMP-010	P1-TBDI-10 (Pin 10)	C1S4.10	29 ms
TMP-JMP-011	P1-TBDI-11 (Pin 11)	C1S4.11	31 ms
TMP-JMP-012	P1-TBDI-12 (Pin 12)	C1S4.12	37 ms
TMP-JMP-013	P1-TBDI-13 (Pin 13)	C1S4.13	28 ms
TMP-JMP-014	P1-TBDI-14 (Pin 14)	C1S4.14	32 ms
TMP-JMP-015	P1-TBDI-15 (Pin 15)	C1S4.15	36 ms
TMP-JMP-016	P1-TBDI-16 (Pin 16)	C1S4.16	24 ms
TMP-JMP-017	P1-TBDI-17 (Pin 19)	C1S4.19	30 ms
TMP-JMP-018	P1-TBDI-18 (Pin 20)	C1S4.20	28 ms
TMP-JMP-019	P1-TBDI-19 (Pin 21)	C1S4.21	27 ms
TMP-JMP-020	P1-TBDI-20 (Pin 22)	C1S4.22	24 ms
TMP-JMP-021	P1-TBDI-21 (Pin 23)	C1S4.23	36 ms
TMP-JMP-022	P1-TBDI-22 (Pin 24)	C1S4.24	39 ms
TMP-JMP-023	P1-TBDI-23 (Pin 25)	C1S4.25	33 ms
TMP-JMP-024	P1-TBDI-24 (Pin 26)	C1S4.26	28 ms
TMP-JMP-025	P1-TBDI-25 (Pin 27)	C1S4.27	37 ms
TMP-JMP-026	P1-TBDI-26 (Pin 28)	C1S4.28	32 ms
TMP-JMP-027	P1-TBDI-27 (Pin 29)	C1S4.29	29 ms

TMP-JMP-028	P1-TBDI-28 (Pin 30)	C1S4.30	42 ms
TMP-JMP-029	P1-TBDI-29 (Pin 31)	C1S4.31	34 ms
TMP-JMP-030	P1-TBDI-30 (Pin 32)	C1S4.32	35 ms
TMP-JMP-031	P1-TBDI-31 (Pin 33)	C1S4.33	28 ms
TMP-JMP-032	P1-TBDI-32 (Pin 34)	C1S4.34	24 ms
TMP-JMP-033	P1-TBDI-33 (Pin 1)	C1S5.1	35 ms
TMP-JMP-034	P1-TBDI-34 (Pin 2)	C1S5.2	36 ms
TMP-JMP-035	P1-TBDI-35 (Pin 3)	C1S5.3	31 ms
TMP-JMP-036	P1-TBDI-36 (Pin 4)	C1S5.4	28 ms
TMP-JMP-037	P1-TBDI-37 (Pin 5)	C1S5.5	25 ms
TMP-JMP-038	P1-TBDI-38 (Pin 6)	C1S5.6	42 ms
TMP-JMP-039	P1-TBDI-39 (Pin 7)	C1S5.7	41 ms
TMP-JMP-040	P1-TBDI-40 (Pin 8)	C1S5.8	32 ms
TMP-JMP-041	P1-TBDI-41 (Pin 9)	C1S5.9	30 ms
TMP-JMP-042	P1-TBDI-42 (Pin 10)	C1S5.10	34 ms
TMP-JMP-043	P1-TBDI-43 (Pin 11)	C1S5.11	34 ms
TMP-JMP-044	P1-TBDI-44 (Pin 12)	C1S5.12	29 ms
TMP-JMP-045	P1-TBDI-45 (Pin 13)	C1S5.13	35 ms
TMP-JMP-046	P1-TBDI-46 (Pin 14)	C1S5.14	24 ms
TMP-JMP-047	P1-TBDI-47 (Pin 15)	C1S5.15	34 ms
TMP-JMP-048	P1-TBDI-48 (Pin 16)	C1S5.16	35 ms
TMP-JMP-049	P1-TBDI-49 (Pin 19)	C1S5.19	22 ms
TMP-JMP-050	P1-TBDI-50 (Pin 20)	C1S5.20	36 ms
TMP-JMP-051	P1-TBDI-51 (Pin 21)	C1S5.21	28 ms
TMP-JMP-052	P1-TBDI-52 (Pin 22)	C1S5.22	39 ms
TMP-JMP-053	P1-TBDI-53 (Pin 23)	C1S5.23	25 ms
TMP-JMP-054	P1-TBDI-54 (Pin 24)	C1S5.24	33 ms
TMP-JMP-055	P1-TBDI-55 (Pin 25)	C1S5.25	22 ms
TMP-JMP-056	P1-TBDI-56 (Pin 26)	C1S5.26	33 ms
TMP-JMP-057	P1-TBDI-57 (Pin 27)	C1S5.27	33 ms
TMP-JMP-058	P1-TBDI-58 (Pin 28)	C1S5.28	22 ms
TMP-JMP-059	P1-TBDI-59 (Pin 29)	C1S5.29	30 ms

TMP-JMP-060	P1-TBDI-60 (Pin 30)	C1S5.30	40 ms
TMP-JMP-061	P1-TBDI-61 (Pin 31)	C1S5.31	26 ms
TMP-JMP-062	P1-TBDI-62 (Pin 32)	C1S5.32	23 ms
TMP-JMP-063	P1-TBDI-63 (Pin 33)	C1S5.33	35 ms
TMP-JMP-064	P1-TBDI-64 (Pin 34)	C1S5.34	27 ms

[illegible]

[illegible]

0 -> 1 -> 0	PASS
0 -> 1 -> 0	PASS
0 -> 1 -> 0	PASS
0 -> 1 -> 0	PASS
0 -> 1 -> 0	PASS

CHASSIS C1 — DIGITAL INPUT DI

Testing C1S6 DI (Channels 1-32) &

Test #	DI Slot & Pin	Target Terminal Tag	Power Source Rail
C1-DIS-001	C1S6.1	P1-TBDI-65 (Pin 1)	TB-24VDC (Rail 1)
C1-DIS-002	C1S6.2	P1-TBDI-66 (Pin 2)	TB-24VDC (Rail 1)
C1-DIS-003	C1S6.3	P1-TBDI-67 (Pin 3)	TB-24VDC (Rail 1)
C1-DIS-004	C1S6.4	P1-TBDI-68 (Pin 4)	TB-24VDC (Rail 1)
C1-DIS-005	C1S6.5	P1-TBDI-69 (Pin 5)	TB-24VDC (Rail 1)
C1-DIS-006	C1S6.6	P1-TBDI-70 (Pin 6)	TB-24VDC (Rail 1)
C1-DIS-007	C1S6.7	P1-TBDI-71 (Pin 7)	TB-24VDC (Rail 1)
C1-DIS-008	C1S6.8	P1-TBDI-72 (Pin 8)	TB-24VDC (Rail 1)
C1-DIS-009	C1S6.9	P1-TBDI-73 (Pin 9)	TB-24VDC (Rail 1)
C1-DIS-010	C1S6.10	P1-TBDI-74 (Pin 10)	TB-24VDC (Rail 1)
C1-DIS-011	C1S6.11	P1-TBDI-75 (Pin 11)	TB-24VDC (Rail 1)
C1-DIS-012	C1S6.12	P1-TBDI-76 (Pin 12)	TB-24VDC (Rail 1)
C1-DIS-013	C1S6.13	P1-TBDI-77 (Pin 13)	TB-24VDC (Rail 1)
C1-DIS-014	C1S6.14	P1-TBDI-78 (Pin 14)	TB-24VDC (Rail 1)
C1-DIS-015	C1S6.15	P1-TBDI-79 (Pin 15)	TB-24VDC (Rail 1)
C1-DIS-016	C1S6.16	P1-TBDI-80 (Pin 16)	TB-24VDC (Rail 1)
C1-DIS-017	C1S6.19	P1-TBDI-81 (Pin 19)	TB-24VDC (Rail 1)
C1-DIS-018	C1S6.20	P1-TBDI-82 (Pin 20)	TB-24VDC (Rail 1)
C1-DIS-019	C1S6.21	P1-TBDI-83 (Pin 21)	TB-24VDC (Rail 1)
C1-DIS-020	C1S6.22	P1-TBDI-84 (Pin 22)	TB-24VDC (Rail 1)
C1-DIS-021	C1S6.23	P1-TBDI-85 (Pin 23)	TB-24VDC (Rail 1)
C1-DIS-022	C1S6.24	P1-TBDI-86 (Pin 24)	TB-24VDC (Rail 1)
C1-DIS-023	C1S6.25	P1-TBDI-87 (Pin 25)	TB-24VDC (Rail 1)
C1-DIS-024	C1S6.26	P1-TBDI-88 (Pin 26)	TB-24VDC (Rail 1)
C1-DIS-025	C1S6.27	P1-TBDI-89 (Pin 27)	TB-24VDC (Rail 1)
C1-DIS-026	C1S6.28	P1-TBDI-90 (Pin 28)	TB-24VDC (Rail 1)
C1-DIS-027	C1S6.29	P1-TBDI-91 (Pin 29)	TB-24VDC (Rail 1)

C1-DIS-028	C1S6.30	P1-TBDI-92 (Pin 30)	TB-24VDC (Rail 1)
C1-DIS-029	C1S6.31	P1-TBDI-93 (Pin 31)	TB-24VDC (Rail 1)
C1-DIS-030	C1S6.32	P1-TBDI-94 (Pin 32)	TB-24VDC (Rail 1)
C1-DIS-031	C1S6.33	P1-TBDI-95 (Pin 33)	TB-24VDC (Rail 1)
C1-DIS-032	C1S6.34	P1-TBDI-96 (Pin 34)	TB-24VDC (Rail 1)
C1-DIS-033	C1S7.1	P1-TBDI-97 (Pin 1)	TB-24VDC (Rail 1)
C1-DIS-034	C1S7.2	P1-TBDI-98 (Pin 2)	TB-24VDC (Rail 1)
C1-DIS-035	C1S7.3	P1-TBDI-99 (Pin 3)	TB-24VDC (Rail 1)
C1-DIS-036	C1S7.4	P1-TBDI-100 (Pin 4)	TB-24VDC (Rail 1)
C1-DIS-037	C1S7.5	P1-TBDI-101 (Pin 5)	TB-24VDC (Rail 1)
C1-DIS-038	C1S7.6	P1-TBDI-102 (Pin 6)	TB-24VDC (Rail 1)
C1-DIS-039	C1S7.7	P1-TBDI-103 (Pin 7)	TB-24VDC (Rail 1)
C1-DIS-040	C1S7.8	P1-TBDI-104 (Pin 8)	TB-24VDC (Rail 1)
C1-DIS-041	C1S7.9	P1-TBDI-105 (Pin 9)	TB-24VDC (Rail 1)
C1-DIS-042	C1S7.10	P1-TBDI-106 (Pin 10)	TB-24VDC (Rail 1)
C1-DIS-043	C1S7.11	P1-TBDI-107 (Pin 11)	TB-24VDC (Rail 1)
C1-DIS-044	C1S7.12	P1-TBDI-108 (Pin 12)	TB-24VDC (Rail 1)
C1-DIS-045	C1S7.13	P1-TBDI-109 (Pin 13)	TB-24VDC (Rail 1)
C1-DIS-046	C1S7.14	P1-TBDI-110 (Pin 14)	TB-24VDC (Rail 1)
C1-DIS-047	C1S7.15	P1-TBDI-111 (Pin 15)	TB-24VDC (Rail 1)
C1-DIS-048	C1S7.16	P1-TBDI-112 (Pin 16)	TB-24VDC (Rail 1)
C1-DIS-049	C1S7.19	P1-TBDI-113 (Pin 19)	TB-24VDC (Rail 1)
C1-DIS-050	C1S7.20	P1-TBDI-114 (Pin 20)	TB-24VDC (Rail 1)
C1-DIS-051	C1S7.21	P1-TBDI-115 (Pin 21)	TB-24VDC (Rail 1)
C1-DIS-052	C1S7.22	P1-TBDI-116 (Pin 22)	TB-24VDC (Rail 1)
C1-DIS-053	C1S7.23	P1-TBDI-117 (Pin 23)	TB-24VDC (Rail 1)
C1-DIS-054	C1S7.24	P1-TBDI-118 (Pin 24)	TB-24VDC (Rail 1)
C1-DIS-055	C1S7.25	P1-TBDI-119 (Pin 25)	TB-24VDC (Rail 1)
C1-DIS-056	C1S7.26	P1-TBDI-120 (Pin 26)	TB-24VDC (Rail 1)
C1-DIS-057	C1S7.27	P1-TBDI-121 (Pin 27)	TB-24VDC (Rail 1)
C1-DIS-058	C1S7.28	P1-TBDI-122 (Pin 28)	TB-24VDC (Rail 1)
C1-DIS-059	C1S7.29	P1-TBDI-123 (Pin 29)	TB-24VDC (Rail 1)

C1-DIS-060	C1S7.30	P1-TBDI-124 (Pin 30)	TB-24VDC (Rail 1)
C1-DIS-061	C1S7.31	P1-TBDI-125 (Pin 31)	TB-24VDC (Rail 1)
C1-DIS-062	C1S7.32	P1-TBDI-126 (Pin 32)	TB-24VDC (Rail 1)
C1-DIS-063	C1S7.33	P1-TBDI-127 (Pin 33)	TB-24VDC (Rail 1)
C1-DIS-064	C1S7.34	P1-TBDI-128 (Pin 34)	TB-24VDC (Rail 1)

RECT DRY CONTACT SIMULATION (64 CHANNELS)

: C1S7 DI (Channels 33-64) via TB-24VDC Toggle Simulation Jig

Simulation Method	Signal Transition	Response Time	PLC Tag Address
16-CH Toggle Switch Box	0 -> 1 -> 0	20 ms	Local:1:I.Data[0]
16-CH Toggle Switch Box	0 -> 1 -> 0	14 ms	Local:1:I.Data[1]
16-CH Toggle Switch Box	0 -> 1 -> 0	16 ms	Local:1:I.Data[2]
16-CH Toggle Switch Box	0 -> 1 -> 0	24 ms	Local:1:I.Data[3]
16-CH Toggle Switch Box	0 -> 1 -> 0	23 ms	Local:1:I.Data[4]
16-CH Toggle Switch Box	0 -> 1 -> 0	27 ms	Local:1:I.Data[5]
16-CH Toggle Switch Box	0 -> 1 -> 0	17 ms	Local:1:I.Data[6]
16-CH Toggle Switch Box	0 -> 1 -> 0	27 ms	Local:1:I.Data[7]
16-CH Toggle Switch Box	0 -> 1 -> 0	28 ms	Local:1:I.Data[8]
16-CH Toggle Switch Box	0 -> 1 -> 0	21 ms	Local:1:I.Data[9]
16-CH Toggle Switch Box	0 -> 1 -> 0	17 ms	Local:1:I.Data[10]
16-CH Toggle Switch Box	0 -> 1 -> 0	28 ms	Local:1:I.Data[11]
16-CH Toggle Switch Box	0 -> 1 -> 0	12 ms	Local:1:I.Data[12]
16-CH Toggle Switch Box	0 -> 1 -> 0	14 ms	Local:1:I.Data[13]
16-CH Toggle Switch Box	0 -> 1 -> 0	14 ms	Local:1:I.Data[14]
16-CH Toggle Switch Box	0 -> 1 -> 0	13 ms	Local:1:I.Data[15]
16-CH Toggle Switch Box	0 -> 1 -> 0	23 ms	Local:1:I.Data[16]
16-CH Toggle Switch Box	0 -> 1 -> 0	26 ms	Local:1:I.Data[17]
16-CH Toggle Switch Box	0 -> 1 -> 0	15 ms	Local:1:I.Data[18]
16-CH Toggle Switch Box	0 -> 1 -> 0	26 ms	Local:1:I.Data[19]
16-CH Toggle Switch Box	0 -> 1 -> 0	16 ms	Local:1:I.Data[20]
16-CH Toggle Switch Box	0 -> 1 -> 0	17 ms	Local:1:I.Data[21]
16-CH Toggle Switch Box	0 -> 1 -> 0	18 ms	Local:1:I.Data[22]
16-CH Toggle Switch Box	0 -> 1 -> 0	26 ms	Local:1:I.Data[23]
16-CH Toggle Switch Box	0 -> 1 -> 0	16 ms	Local:1:I.Data[24]
16-CH Toggle Switch Box	0 -> 1 -> 0	14 ms	Local:1:I.Data[25]
16-CH Toggle Switch Box	0 -> 1 -> 0	16 ms	Local:1:I.Data[26]

16-CH Toggle Switch Box	0 -> 1 -> 0	21 ms	Local:1:I.Data[27]
16-CH Toggle Switch Box	0 -> 1 -> 0	22 ms	Local:1:I.Data[28]
16-CH Toggle Switch Box	0 -> 1 -> 0	24 ms	Local:1:I.Data[29]
16-CH Toggle Switch Box	0 -> 1 -> 0	14 ms	Local:1:I.Data[30]
16-CH Toggle Switch Box	0 -> 1 -> 0	13 ms	Local:1:I.Data[31]
16-CH Toggle Switch Box	0 -> 1 -> 0	23 ms	Local:2:I.Data[0]
16-CH Toggle Switch Box	0 -> 1 -> 0	26 ms	Local:2:I.Data[1]
16-CH Toggle Switch Box	0 -> 1 -> 0	13 ms	Local:2:I.Data[2]
16-CH Toggle Switch Box	0 -> 1 -> 0	18 ms	Local:2:I.Data[3]
16-CH Toggle Switch Box	0 -> 1 -> 0	20 ms	Local:2:I.Data[4]
16-CH Toggle Switch Box	0 -> 1 -> 0	22 ms	Local:2:I.Data[5]
16-CH Toggle Switch Box	0 -> 1 -> 0	14 ms	Local:2:I.Data[6]
16-CH Toggle Switch Box	0 -> 1 -> 0	20 ms	Local:2:I.Data[7]
16-CH Toggle Switch Box	0 -> 1 -> 0	16 ms	Local:2:I.Data[8]
16-CH Toggle Switch Box	0 -> 1 -> 0	20 ms	Local:2:I.Data[9]
16-CH Toggle Switch Box	0 -> 1 -> 0	26 ms	Local:2:I.Data[10]
16-CH Toggle Switch Box	0 -> 1 -> 0	18 ms	Local:2:I.Data[11]
16-CH Toggle Switch Box	0 -> 1 -> 0	25 ms	Local:2:I.Data[12]
16-CH Toggle Switch Box	0 -> 1 -> 0	15 ms	Local:2:I.Data[13]
16-CH Toggle Switch Box	0 -> 1 -> 0	28 ms	Local:2:I.Data[14]
16-CH Toggle Switch Box	0 -> 1 -> 0	22 ms	Local:2:I.Data[15]
16-CH Toggle Switch Box	0 -> 1 -> 0	19 ms	Local:2:I.Data[16]
16-CH Toggle Switch Box	0 -> 1 -> 0	15 ms	Local:2:I.Data[17]
16-CH Toggle Switch Box	0 -> 1 -> 0	20 ms	Local:2:I.Data[18]
16-CH Toggle Switch Box	0 -> 1 -> 0	17 ms	Local:2:I.Data[19]
16-CH Toggle Switch Box	0 -> 1 -> 0	12 ms	Local:2:I.Data[20]
16-CH Toggle Switch Box	0 -> 1 -> 0	12 ms	Local:2:I.Data[21]
16-CH Toggle Switch Box	0 -> 1 -> 0	24 ms	Local:2:I.Data[22]
16-CH Toggle Switch Box	0 -> 1 -> 0	26 ms	Local:2:I.Data[23]
16-CH Toggle Switch Box	0 -> 1 -> 0	13 ms	Local:2:I.Data[24]
16-CH Toggle Switch Box	0 -> 1 -> 0	22 ms	Local:2:I.Data[25]
16-CH Toggle Switch Box	0 -> 1 -> 0	19 ms	Local:2:I.Data[26]

16-CH Toggle Switch Box	0 -> 1 -> 0	20 ms	Local:2:I.Data[27]
16-CH Toggle Switch Box	0 -> 1 -> 0	27 ms	Local:2:I.Data[28]
16-CH Toggle Switch Box	0 -> 1 -> 0	18 ms	Local:2:I.Data[29]
16-CH Toggle Switch Box	0 -> 1 -> 0	22 ms	Local:2:I.Data[30]
16-CH Toggle Switch Box	0 -> 1 -> 0	22 ms	Local:2:I.Data[31]

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[illegible]

PASS
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PASS

CHASSIS C1 DI/DO FAT ACCEPTANCE

Project Name:	Ingredion Thailand — SPRINT 18K TPA Spray Dryer Plant (Kalasin)
Target Control System:	Rockwell Automation ControlLogix 5580 — Chassis C1 (Main CPU & Core I/O)
Test Scope:	100% Verification of 64 DO Channels (C1S8, C1S9) & 128 DI Channels (C1S4, C1S5)
Test Standards:	IEC 62381 / ISA-RP60.8 / NFPA 79 Electrical Safety Standards
Testing Venue:	xDev Engineering Workshop, Assembly & Integration Testing Bay 1
Test Date:	08-October-2026
Overall Pass Rate:	192 / 192 Channels Tested (100.0% PASS Rate 0 Open Category A Defects)
Final Verdict:	CHASSIS C1 DI/DO WIRING & HARDWARE OFFICIALLY ACCEPTED

REPRESENTATIVE ENDORSEMENTS:

Client Lead Representative

Ingredion (Thailand) Co., Ltd.
Project Engineering Manager
Date: ____ / ____ / 2026

Lead Auditor

Kalasin Erwan
Senior Automation Engineer
Date: ____

ANCE & ENDORSEMENT CERTIFICATE

5, C1S6, C1S7)

ts)

ED FOR COMMISSIONING

mation Integrator

gineering Co., Ltd.

tomation Engineer

___ / ___ / 2026

Quality & Safety Officer

xDev Workshop Integration Center

QA/QC Compliance Lead

Date: ___ / ___ / 2026